START: A Spatiotemporal Autoregressive Transformer for Enhancing Crime Prediction Accuracy

Umair Muneer Butt, Sukumar Letchmunan, Mubashir Ali, and Hafiz Husnain Raza Sherazi

***Abstract*—This study presents a spatiotemporal autoregressive transformer (START) that combines the strengths of spatiotemporal transformers and vector autoregression (VAR) to predict crime by type, handling nonstationary data and capturing long-term dependencies. Moreover, Start utilizes one-hot encoding to differentiate crime types, enabling the model to assign specific attention to individual crime categories. Seasonal trend decomposition using LOESS (STL) is employed to decompose time series into trend, seasonality, and remainder components by effectively capturing residual variations. VAR attention enhances prediction accuracy by focusing on specific crime types and prioritizing relevant information. The proposed approach is evaluated on publicly available crime datasets from New York, Chicago, and Los Angeles and three state-of-the-art evaluation measures: mean error (ME), mean absolute percentage error (MAPE), and root mean square error (RMSE). Our simulation results demonstrate the significance of the proposed method in enhancing crime prediction accuracy for the most critical crimes compared to state-of-the-art methods. Enhanced crime prediction may help law enforcement agencies allocate resources efficiently by focusing on the most critical areas.**

***Index Terms*—Crime prediction, safe society, spatiotemporal, time series analysis, transformers.**

I. Introduction

CRIME is a social drawback, undermining the standard of everyone's life and affecting the developmental growth rate and the economic progress of any society [1], [2]. Considering the crime trends of society, individuals may face significant challenges when deciding whether to move to a new town or which places to avoid while traveling [3], [4].

Traditional crime prediction approaches such as autoregressive integrated moving averages (ARIMAs) models, and exponential smoothing (ES) have been widely used [5], [6], [7] for short-term forecasting. However, these methods assume stationarity in the data and are limited in their ability to handle complex nonlinear trends, multivariate relationships, and seasonal variations. Conversely, vector autoregression (VAR) is suitable for multivariate analysis and can handle stationary and nonstationary data by using lagged values to capture trends for short-term forecasting [8], [9]. However, VAR models are limited in handling complex nonlinear relationships and long-term dependencies [10].

One possible solution is deep learning models such as long short-term memory (LSTM) [11], bidirectional LSTM (BiLSTM) [12], and spatiotemporal neural networks (STNNs) [13] are capable of handling complex nonlinear, intricate patterns, long-term dependencies, and can provide generalized representations. However, these traditional deep-learning models primarily focus on temporal dependencies, lacking in capturing geographic distributions. Moreover, they struggle to capture long-term dependencies over extended periods, such as annual trends, due to vanishing gradients [14], [15]. Moreover, there is a lack of research identifying crime according to crime types and the seasonality of crime. Most attempts have focused on crime number predictions [16], [17]. However, various studies have enlightened the significance of crime prediction by crime type.

Recently, spatiotemporal transformers have advanced significantly, enabling them to model spatial and temporal relationships in the data [18]. They can capture long-term dependencies over longer horizons using attention mechanisms [19], [20]. The evolution of transformers has led to recent breakthroughs in time-series analysis, tracking back to their introduction in 2017 [21]. The transformers currently emphasize natural language processing (NLP) tasks, but their architecture has recently been extended to encompass categorization, time-series analysis, semantic segmentation, and other tasks [22]. Transformers were not initially designed to handle time series data. Researchers have also tuned and improved the architectures to improve the efficiency of time-series processing [23], [24]. However, transformers struggle to find interdependencies in crime types and nonstationary trends and patterns of data. Therefore, there is a need to develop enhanced spatiotemporal transformers that can focus on specific crime types and can handle both stationary and nonstationary data.

Hybrid models combining the strengths of deep learning and statistical approaches have also been proposed in the literature for time series forecasting. Umair et al. [25] study introduced a hybrid of BiLSTM and ES for crime forecasting. The proposed

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method is assessed using New York crime data from 2010 to 2017. The proposed method is more efficient than state-of-the-art seasonal autoregressive integrated moving averages (SARIMA) with a low MAPE, RMSE, and MAE. Thus, the suggested approach may aid law enforcement agencies in crime prevention and control by forecasting crime patterns. Therefore, this study proposes a spatiotemporal transformer and VAR as an attention mechanism to focus on specific crime types to enhance crime prediction accuracy.

II. Related Work

This section covers recent advances in spatiotemporal crime prediction, including crime hotspot prediction techniques, their advantages, challenges, and a summary of state-of-the-art data mining and machine learning approaches used for crime analysis.

A. Background

Crime prediction is a critical research area aimed at ensuring public safety globally. Accurate crime prediction enables law enforcement agencies to allocate resources effectively and helps prevent criminal activities [26]. Existing crime prediction methods predominantly rely on spatial statistics and basic time series analysis techniques [27], [28], [29]. While these methods provide valuable insights, they often fail to capture complex interrelations between spatial and temporal dynamics.

Deep learning models have demonstrated significant promise in understanding intricate patterns in spatiotemporal data. Recurrent neural networks (RNNs), convolutional neural networks (CNNs), and transformers have excelled at learning and representing such relationships with notable accuracy. The inclusion of attention mechanisms in deep learning models, such as LSTM and transformers, has further enhanced their ability to focus on relevant data [30], [31]. These mechanisms provide insights into future crime hotspots by emphasizing specific regions and timestamps.

However, relying solely on historical crime data can overlook broader external factors, such as seasonality and economic fluctuations, which significantly influence crime trends [32], [33]. Traditional analyses often centered on univariate approaches such as ARIMA and ES variants, fail to account for the variations in crime trends across different crime types. Advanced frameworks, such as VAR with exogenous variables, address these complexities by enabling multidimensional predictions [34], [35]. By integrating these methodologies, this research aims to develop a robust approach to improve the accuracy and reliability of crime predictions, aiding law enforcement in proactive crime prevention strategies.

B. State-of-the-Art Approaches for Crime Prediction

The limitations of traditional time series forecasting models in capturing complex spatiotemporal dependencies render them less effective for tasks such as crime prediction. In contrast, spatiotemporal transformers have emerged as a breakthrough in this field [36], [37]. These deep learning architectures excel at analyzing spatial and temporal patterns simultaneously, making them suitable for applications involving time-dependent and spatially distributed data.

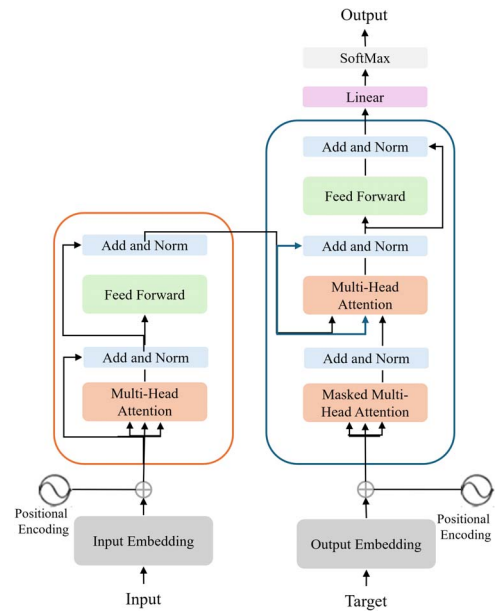

Fig. 1. State-of-the-art architecture of a transformer.

1) Recent Advances on Transformers: Spatiotemporal transformers leverage the encoder–decoder architecture of traditional transformers, incorporating spatial information. The encoder extracts spatial and temporal features using self-attention mechanisms, enabling the model to identify relationships between data points at both spatial and temporal levels. The decoder utilizes this information to produce predictions that account for both spatial and temporal dynamics, as illustrated in Fig. 1.

Recent advancements in spatiotemporal transformers include the following.

Graph Neural Networks (GNNs): Integration with GNNs helps analyze complex spatial relationships, such as those found in social networks and transportation systems [38].

Attention Mechanisms: Enhanced attention mechanisms emphasize specific spatial and temporal regions, improving reliability and interpretability [39].

Multiscale Modeling: Incorporating multiscale network structures enables the recognition of patterns at various spatial and temporal resolutions [40].

The Standard Transformer, embodied in the encoder–decoder architecture with the self-attention mechanisms, deals well with the long-range dependencies within sequential data. Furthermore, numerous architectural designs are suggested in the literature for diverse purposes. Vision transformer is a variant that employs transformer architecture for image processing that is compatible with incorporating spatial information from crime location maps [41]. Bidirectional encoder representations from transformers (BERT) is a pretrained large text corpus (dataset) that can be used for the analysis of crime-related texts such as news articles and social media posts [42]. Graph transformer utilizes GNNs to examine relational data such as social

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networks, which might help disclose criminal behavior patterns. Spatiotemporal transformers (ST-Transformers) are a type of transformer architecture that is designed especially for spatiotemporal data, including spatial convolutions and temporal self-attention to incorporate both spatial and temporal aspects of crime data [37].

Shi et al. [43] proposed TSE-Tran, a hybrid model combining temporal representation and transformer architectures for telecommunication fraud prediction. While effective, its feature set was limited to incident time series data. Similarly, Yag et al. [44] introduced CrimeTrans, a transformer-based model for analyzing textual data in criminal hotspots. Despite its innovative use of textual features, this model's performance was constrained by data quality issues.

Although transformers excel at capturing long-range dependencies and complex spatiotemporal relationships, they face challenges such as computational complexity and interpretability. Additionally, their reliance on extensive data can hinder applicability in certain scenarios [45]

Transformers and their adaptations are good at handling significant long-range and composite spatiotemporal relationships within data. Moreover, it can use different data sources, such as images, text, and spatial data. Furthermore, they can be pretrained on large datasets for better performance. Conversely, transformers can be computationally burdensome, especially for the complex versions. In addition, interpretability is another issue that makes it hard to reveal the model's decision-making process. Therefore, the models need huge amounts of data, which is hard to collect in some application areas [45], [46].

2) VAR: A Complementary Approach: VAR models are statistical techniques designed to capture interdependencies among multiple time series variables, making them suitable for short-term forecasting [47]. Variants such as VARX (with exogenous variable), Bayesian VAR (handling prior information), and time-varying autoregression (TVAR) adapt to evolving patterns in crime data [48], [49], [50].

Yadav et al. [51] demonstrated VAR's ability to capture temporal dependencies among different crime categories, enabling accurate short-term predictions. Freilich et al. [52] utilized VAR to identify spatiotemporal patterns in crime data. However, these studies often focus on average crime rates, neglecting the heterogeneity of crime types and long-term trends.

Cheng et al. [53] designed a short-term traffic flow forecasting system. The VAR model derived from econometric theory and the CNN-LSTM hybrid neural network model obtained from deep learning are used in the analysis. The VAR model is applied first to evaluate an inherent relationship between traffic variables, and the predictable relationship of these variables is obtained. Next, the CNN-LSTM model is adopted to predict the multifeature speed for one spatial location, and the results illustrate that the multifeature prediction is better than the single-feature prediction. After that, numerous popular deep learning models, as well as other shallow predicted models, are proposed to be compared with the designed CNN-LSTM network model, and the comparison presents that the model performance of the constructed CNN-LSTM network model is greater compared to other models in predicting the short-term traffic flow. Nevertheless, the article broadly examines the development of the hybrid model without sufficient performance testing on actual crime datasets. In contrast, the computational complexity of the hybrid approach and its feasibility in crime prediction for large-scale scenarios have not been given enough attention.

VAR is a state-of-the-art method with straightforward interpretability, and the model variables are easy to understand. VAR has been used in the literature for multivariate analysis individually and as a hybrid with deep learning techniques for capturing more dynamic complexities. They generally perform better in the short term and need less data than the deep learning models. Nevertheless, their lack of long-range dependencies and complex spatial patterns renders them ineffective for complex tasks. In addition, they mainly concentrated on short-term predictions that were not accurate for long periods. Moreover, they might not cope with the shifting crime patterns effectively.

III. Proposed Methodology

This study introduces the spatiotemporal autoregressive transformer (START), a hybrid model leveraging the strength of state-of-the-art spatiotemporal transformers (ST-Transformers) and VAR to enhance crime prediction accuracy. The proposed approach predicts crime occurrences by type, focusing on spatial and temporal dependencies within crime data.

The novelty of START lies in combining the complementary capabilities of ST-Transformers and VAR to capture intricate spatial relationships on a spatial grid while modeling temporal dynamics within each location. By learning long-range dependencies and adapting to nonstationary behavior in the data, the model addresses limitations in simpler approaches.

Fig. 2 illustrates the proposed START method architecture. The design captures the intricate spatial relationships and evolving temporal dynamics in crime datasets. The architecture consists of several key components. First, the STL block is pivotal in extracting seasonal and trend components from the raw time series. The detrended and deseasonalized residuals form the basis for further analysis. These preprocessing steps ensure the model adapts to nonstationary behaviors in the data. A key innovation in START is the multihead VAR attention mechanism, which enhances temporal modeling by learning dependencies across multiple time series. This module combines linear operations, differencing, and concatenation to form a representation that captures short-term variations and cross-series interactions.

Moreover, the encoder module employs a multilevel stacking approach where norm layers and frequency attention refine the representation at each level. The residuals are iteratively processed, enriching the representation with latent seasonal and trend patterns across levels. The ST-encoder captures spatiotemporal patterns by iteratively embedding spatial and temporal relationships across multiple layers. It relies on one-hot encoding and learned embeddings to represent the spatiotemporal context effectively.

The ST-decoder reconstructs the temporal sequence by stacking seasonal (S), trend (T), and level (L) components extracted by the encoder. Each stack contributes to the final prediction by incorporating both recent patterns and long-term trends. Fig. 3

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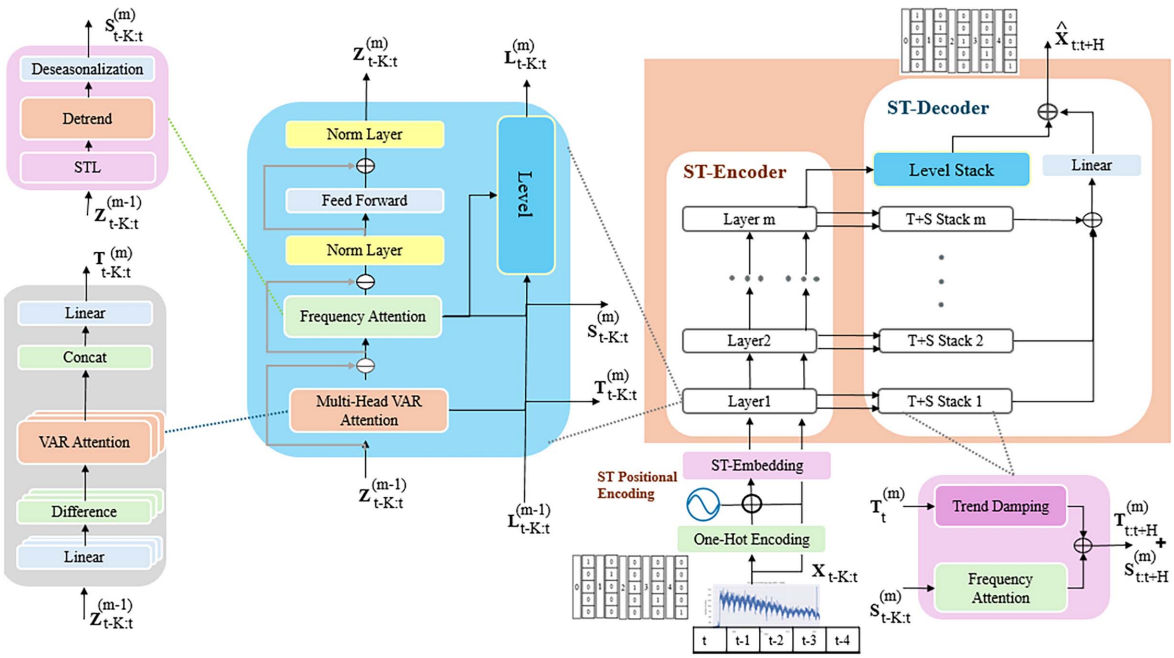

Fig. 2. Proposed architecture of START for crime prediction: START integrates spatiotemporal transformers and multihead VAR attention with STL decomposition to capture spatial relationships, temporal dynamics, and nonstationary behaviors, producing accurate crime forecasts.

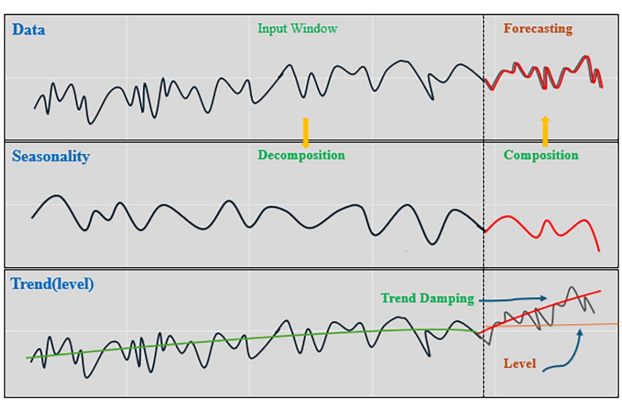

Fig. 3. Decomposition of input into seasonal and trend components.

presents the START mechanism, which performs a similar function by isolating trend and seasonal components from the intermediate representation for forecasting. The decomposed trend component, with growth and level effects, facilitates predictions with controlled extrapolation. The architecture combines output from seasonal, trend, and level components using a linear layer to produce the final forecast.

A. Problem Formulation

Let $x_t \in \mathbb{R}^m$ represents a multivariate time series crime observation at the timestamp t. Here, m is the dimensionality of the data. Given a sequence of past observations $X_{t-K:t} = [x_{t-K}, \ldots, x_{t-1}]$ from timestamp $t-K$ to $t-1$ represents input time series data of crime of length K. The aim is to predict future values over a horizon $X_{t:H} = [x_t, \ldots, x_{t+H-1}]$, where $\hat{X}_{t:t+H}$ shows the point forecast of the future values over the horizon. It is a vector of size $H*m$, and each row represents the predicted value for each type at a future time stamp. Therefore, the objective is to establish a forecasting function $\hat{X}_{t:t+H} = f(X_{t-K:t})$ by minimizing some loss function.

B. Spatiotemporal Embeddings (ST-Embedding)

This section discusses the necessary data preparation steps to transform the raw data into a suitable form. Crime dataset $\mathbf{D}$ with the date column is converted into a suitable format to filter out invalid entries such as $\mathbf{D}_{\text{valid}} = \{d \in \mathbf{D} \mid \text{is_valid_date}(d_{\text{Date}})\}$. Moreover, the top five most frequent crime types $\mathbf{C}$ are extracted based on their crime count, such as $\mathbf{C}_{\text{top5}} = \{\text{top 5 crimes from } \mathbf{C} \mid \text{count}(c)\}$. After finding the five most frequent crime types, the $\mathbf{D}_{\text{valid}}$ data are filtered further to retain only the selected crime types $\mathbf{D}_{\text{filtered}} = \{d \in \mathbf{D}_{\text{valid}} \mid d_{\text{Primary Type}} \in \mathbf{C}_{\text{top5}}\}$ to reduce complexity by focusing on the most relevant data. In addition, one hot encoding extracts hours and location information from the processed data into one hot encoded vector, as shown in (1). The $\oplus$ shows the concatenation operation of spatial and temporal features

$$\mathbf{H}_{\text{onehot}} = \text{OneHotEncode}\{\mathbf{h} \mid \mathbf{h} = \text{hour}(\mathbf{d}_{\text{Date}}) \oplus (l \mid l = d_{\text{Location Description}}), d \in \mathbf{D}_{\text{filtered}}\}. \quad (1)$$

Furthermore, positional encoding is applied to the spatial and temporal features, providing information about the position of each element in the sequence. This helps the model to understand the order of data points that are significant for time series analysis, as shown in

$$\mathbf{P}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10\,000^{2i/d}}\right)$$
$$\mathbf{P}(\text{pos}, 2i+1) = \cos\left(\frac{\text{pos}}{10\,000^{2i/d}}\right). \quad (2)$$

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C. Spatiotemporal Encoder (ST-Encoder)

The spatial–temporal encoder (ST-Encoder) is the core of START, consists of m encoder layers, and applies embedding and positional encoding to the input. Embeddings convert input tokens into dense vectors, and positional encoding adds information about the positions of the tokens in the sequence. These ST-encoder layers comprised of multihead, frequency, and VAR attention, followed by a feed-forward network (FFN), layer normalization, and dropout.

While the ST-encoder encodes a representation of sinusoidal trends and seasonality from a configurable window, the decoder predicts the output time series. Given this objective, the classical methods are mainly based on supposing seasonality characteristics, which can be additives or multiplicative [54], [55]. With residual learning from [56] and [57], this study develops a deep architecture accommodating topographically intrinsic patterns and solves the vanishing gradient problem by improving training efficiency. The following section discusses the core components of the ST-encoder within each layer.

1) Multihead Attention (MHA): The MHA mechanism consists of linear projections, scale dot product attention, multihead attention, dropout, and residual connections. MHA allows the model to attend different positions of the input sequence simultaneously by capturing various relationships and dependencies within the data. Input to the MHA is tensor $\mathbf{X} \in \mathbb{R}^{N \times T \times d_{\text{model}}}$, where N is the batch size, T is the sequence length, and d_{model} is the dimensionality of the model. The input is then projected to query $(\mathbf{Q})$, key $(\mathbf{K})$, and value $(\mathbf{V})$ matrices using learnable weight matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$, as shown in

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V. \tag{3}$$

In addition, a scaled dot product attention is computed using query, key, and value scaled by $\sqrt{d_k}$ that shows the dimensionality of the keys and applies softmax function to obtain attention weights, as shown in

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}. \tag{4}$$

The multiple heads allow the model to focus on different sequence parts. The heads' output is then concatenated and linearly transformed, as shown in 5. A head in the attention mechanism is computed as: $\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_Q^i, \mathbf{K}\mathbf{W}_K^i, \mathbf{V}\mathbf{W}_V^i)$ and $\mathbf{W}_O$ is a learnable weight matrix

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}_O. \tag{5}$$

Finally, dropout is applied to attention to prevent overfitting. Moreover, the output of the MHA layer is added to the original input as residual learning, followed by layer normalization to stabilize and speed up the training process, as shown in

$$\text{out1} = \text{LN}(\mathbf{X} + \text{Dropout}(\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}))). \tag{6}$$

2) Frequency Attention: Frequency attention decomposes the input (X) into seasonal (S) and trend (T) components and calculates an attention score based on these components to capture intrinsic periodic patterns in the data. Moreover, a hyperbolic tangent activation function is utilized to get an intermediate representation $\mathbf{h}$, as shown in

$$\begin{aligned} \mathbf{h} &= \tanh(\mathbf{W}_s\mathbf{x}_s + \mathbf{W}_t\mathbf{x}_t) \\ \mathbf{c} &= \sum_i \alpha_i \mathbf{x}_{s,i} \\ \text{Output}_{\text{freq}} &= \text{MultiHead}(Q, K, V) + c_t. \end{aligned} \tag{7}$$

The score $\mathbf{s}$ is calculated by projecting $\mathbf{h}$ using a learnable weight matrix $\mathbf{W}_a$ such as $\mathbf{s} = \mathbf{W}_a\mathbf{h}$. The attention weights (α_i) are computed using a softmax function over the scores: $\alpha_i = (\exp(s_i)/\sum_j \exp(s_j))$. Furthermore, the context vector is calculated using a weighted sum of seasonal components. Finally, the context vector $\mathbf{c}$ is added to the output of the multihead-attention layer.

3) Multihead VAR Attention (MH-VARA): Multihead VAR attention captures temporal dependencies and patterns by learning residuals with VAR. A VAR model is then estimated based on the residual, given the window of attention outputs $\mathbf{X}_{\text{window}} = [\mathbf{x}_{t-n}, \ldots, \mathbf{x}_t]$. The residuals in VAR are computed as shown in

$$\begin{aligned} y_t &= A_1 y_{t-1} + A_2 y_{t-2} + \cdots + A_p y_{t-p} + \epsilon_t \\ \epsilon_t &= y_t - \sum_{i=1}^{p} A_i y_{t-i} \\ \text{Output}_{\text{VAR}} &= \text{MultiHead}(Q, K, V) + \epsilon_t. \end{aligned} \tag{8}$$

Finally, the VAR context is reshaped and padded to match the output dimensions (out1+ = var_context_padded).

4) Feed Forward Network (FFN): The FFN provides a significant nonlinearity and model capacity by applying two nonlinear transformations with Relu activation functions between them, as shown in

$$\text{FFN}(\mathbf{X}) = \text{ReLU}(\mathbf{X}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2. \tag{9}$$

Every output of the previous layer passes through FFN, and dropout is applied to the FFN output. The processed output is then added to the input of the FFN layer, followed by layer normalization, as shown in

$$\text{output} = \text{LN}(\text{out1} + \text{Dropout}(\text{FFN}(\text{out1}))). \tag{10}$$

D. Spatiotemporal Decoder (ST-Decoder)

The spatiotemporal decoder (ST-Decoder) produces a k-step ahead forecast. The ST-decoder block utilizes M copies of a stacked structure comprised of Trend and seasonality (T + S) and a level component. ST-decoder processes and generates sequences by leveraging the encoded representations from the encoder. This sophisticated architecture enables the model to predict future crime incidents by understanding the patterns and dependencies within the historical data. The ST-decoder consists of several key components, such as m decoder layers, embedding layer, and positional encoding. First, the self-attention mechanism in the decoder allows each position in the target sequence to attend to all previous positions, including

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the current position and encoder–decoder attention. A look-ahead mask ensures that each position only attends to previous positions.

The embedding layer maps target sequence indices (representing previous crime counts) into high-level dense vectors ($E = \text{Embedding}(x) \times \text{units} + \text{PositionalEncoding}(x)$). This conversion is very important for identifying the semantic meaning of each token. The positional encoding adds a signal to the model about the location of the tokens in the sequence, which is important for capturing temporal dependencies in the crime data. Moreover, the self-attention mechanism allows the model to attend to different positions in the target sequence itself, which helps to capture dependencies within the sequence, such as trends in crime counts, as shown in

$$\text{attn}_1 = \text{LN}\left(x + \text{Dropout}\left(\text{softmax}\left(\frac{Q_1 K_1^T}{\sqrt{d_k}}\right) V_1\right)\right). \quad (11)$$

In addition, the encoder–decoder attention allows the decoder to focus on relevant parts of the encoder's output. It incorporates context from the input sequence to generate the output, as shown in

$$\text{attn}_2 = \text{LN}\left(\text{attn}_1 + \text{Dropout}\left(\text{softmax}\left(\frac{Q_2 K_2^T}{\sqrt{d_k}}\right) V_2\right)\right). \quad (12)$$

This initialization also incorporates the dropout layers when training to avoid over-training. The main computations are performed in the stacked instances of the decoder layer. Each layer has several key components: two multihead attention mechanisms, a FFN, and other stacks for processing time-related features. Moreover, additional stacks are included within each decoder layer to improve the model's temporal complexity in the crime data. They include trend damping, level stacking, and frequency attention mechanisms. Trend damping is useful in accounting for long-term trends in the crime rate data, and level stacking concentrates on the total amount of crime in a given period or area, as shown in (13). Conversely, frequency attention is intended to find periodic patterns in the data, such as daily and weekly fluctuations in criminal incidents

$$\begin{aligned} \text{trend_damping_output} &= W_{\text{trend}} \cdot \text{attn}_2 + b_{\text{trend}} \\ \text{level_stack_output} &= W_{\text{level}} \cdot \text{attn}_2 + b_{\text{level}} \\ \text{freq_attention_output} &= \text{FrequencyAttention}(\text{ attn}_2). \end{aligned} \quad (13)$$

The outputs that have gone through the attention mechanisms are concatenated (combined_output = trend_damping_output + level_stack_output + freq_attention) before being fed to a FFN with two fully connected layers separated by a ReLU activation function. This network introduces nonlinearity into the model and makes it possible for the model to capture different nonlinear patterns in the data. The final output from the FFN takes normalization and dropout to make it more stable and reduce the risk of overfitting, as shown in

$$\begin{aligned} \text{ffn_output} = \text{LN}(\text{attn}_2 + \ \text{Dropout}(\text{ReLU}(W_1 \cdot \\ \text{combined_output} + b_1) W_2 + b_2)). \end{aligned} \quad (14)$$

The output produced through each decoder layer is successively updated as it recurs through several decoder layers. The final output from the last decoder layer is the model's prediction for the future time steps in a given sequence. In the context of crime prediction, this output would mean the number of each type of crime predicted to occur at each time step in the future.

E. Frequency Attention

The frequency layer is the crucial component of the model designed to enhance the ability to capture and focus on periodic patterns in the time series data. It decomposes the input time series data into seasonality, trend, and residual components, which are crucial for understanding underlying temporal patterns. Primarily, this study uses STL decomposition on the input data to split it into seasonal, trend, level, and residual components, as shown in

$$\begin{aligned} x(t) &= s(t) + t(t) + r(t) \\ \text{level} &= \frac{1}{n} \sum_{i=1}^{n} t(i) \end{aligned} \quad (15)$$

where $s(t)$ is the seasonal, $t(t)$ is trend, and $r(t)$ is residual component. The level component is calculated as the mean of the trend component. Moreover, an attention mechanism is used within the layer to selectively on seasonal and trend components. It calculates attention scores for each sequence based on seasonal and trend components. Moreover, the softmax function converts the scores into attention weights, and context vectors are formulated, as shown in

$$\begin{aligned} \text{score} &= V\left(\tanh\left(W(s) + U(t)\right)\right) \\ \alpha &= \text{softmax}(\text{score}) \\ c &= \sum_{i=1}^{n} \alpha_i s_i \\ c &= \sum_{i=1}^{\text{seq_len}} \alpha_i s_i. \end{aligned} \quad (16)$$

Generally, each sample and feature in the batch converts feature data into a numpy array. STL decomposition is performed to obtain seasonal, trend, level, and residual components. Moreover, detrend (detrended $= x(t) - t(t)$) and deseasonalized (deseasonalized $= x(t) - s(t)$) data are also extracted to understand patterns efficiently by removing seasonality and trends from the data. Frequency attention output context vectors, attention weights, level, detrend, and deseasonalized data are used within the transformer encoder and decoder to efficiently enhance the model's ability to capture periodic patterns.

F. Multihead Vector Autoregression Attention (MH-VARA)

VAR is an excellent method of analyzing multivariate time series data. It describes the time-varying interrelatedness of different variables. However, classical VAR models do not consider nonlinear connections, and lags are used. This analysis employs an attention mechanism as a part of the VAR framework (VARA) to eliminate these shortcomings. The attention part allows the model to concentrate on each variable's most important past values to predict the future value of another

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one. It weighs different lagged values based on their relevance in forecasting the target variable. This facilitates the model in discovering intricate dependencies between different variables. We emphasized the importance of attention weights in the VAR model in predicting future outcomes. Rather than a weighted sum of lagged values, the model takes a weighted sum from attendee scores. Therefore, the model can concentrate on each variable's most relevant old values while forecasting the future.

Primarily, VARA is designed to capture and model time series dependencies between five types of crimes by combining the multihead attention mechanism with VAR. This layer helps the model handle complex temporal patterns and dependencies, leading to more accurate and robust predictions. The attention mechanism allows the model to focus on different parts of the input sequence by capturing short-term dependencies and interaction between different crime-type features. The VAR approach captures long-term dependencies and trends in the data by modeling the relationship between different features over time. This is very beneficial in time series analysis when the current state is influenced by previous layers in the model.

The multihead attention mechanism provided parallel processing power so that the model could process different parts of the input sequence parallel. It enables the model to capture diverse sets of dependencies and patterns. Moreover, each head can focus on different aspects of the data, leading to a more comprehensive understanding of the features interactions. Finally, the residual analysis from the VAR model provides insight into the parts of data that are not explained by linear relationships, allowing the model to capture more complex nonlinear dependencies.

Input to the MH-VARA is the number of heads, units, depth, max_lag fo VAR, and window size is the window size for VAR analysis. The multihead mechanism splits the input into separate attention heads $x \in \mathbb{R}^{\text{batch_size} \times \text{seq_len} \times \text{units}} \rightarrow x \in \mathbb{R}^{\text{batch_size} \times \text{seq_len} \times \text{num_heads} \times \text{depth}}$. Moreover, it computes query $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ using dense layers and splitting them into heads, as shown in

$$\begin{aligned} \mathbf{Q} &= \text{split_heads}(\mathbf{W_q}(\mathbf{q}), \text{batch_size}) \\ \mathbf{K} &= \text{split_heads}(\mathbf{W_k}(\mathbf{k}), \text{batch_size}) \\ \mathbf{V} &= \text{split_heads}(\mathbf{W_v}(\mathbf{v}), \text{batch_size}). \end{aligned} \tag{17}$$

In addition, a scaled dot product attention is computed using query, key, and value scaled by $\sqrt{d_k}$ that shows the dimensionality of the keys and applies the softmax function to obtain attention weights, as shown in

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}. \tag{18}$$

For VAR modeling, this study utilized a sliding window approach to extract segments of the attention output. VAR is then fitted on each segment, and residuals are stored, as shown in

$$\begin{aligned} \text{window_data} &= \text{concat_attention_np}[i, j : j + \text{window_size}, :] \\ \text{var_model} &= \text{VAR}(\text{window_data}) \\ \text{var_result} &= \text{var_model.fit}(\text{max_lag}). \end{aligned} \tag{19}$$

TABLE I
CRIME DATASETS SPECIFICATIONS

Country	Population	Attributes	Records	Categories
Chicago	2.7M	22	7.2M	28
New York	8.4M	35	2.1M	30
Los Angeles	3.8M	28	2.5M	17

Subsequent layers use the context vector and enriched representations in the transformer model to generate predictions.

IV. EXPERIMENTAL EVALUATION

The proposed START methodology is evaluated through rigorous experiments using publicly available spatiotemporal crime datasets from New York City, Chicago, and Los Angeles [58], [59], [60]. These datasets contain spatial and temporal information essential for analyzing and forecasting crimes. The evaluation aims to assess the effectiveness of the proposed model in accurately predicting the frequency and type of crimes. Furthermore, the model's performance is compared with state-of-the-art approaches to validate its superiority.

A. Dataset

The study uses crime data spanning 2010–2017 with key attributes such as date, crime type, geographic coordinates (latitude and longitude), and crime description. The dataset details are summarized in Table I.

For each city, the data were split into training (80%) and testing (20%) sets. Training data from 2010 to 2013 are used to forecast crime occurrences for 2014, with subsequent sliding windows applied for additional predictions [61], [62], [63].

B. Data Preprocessing

Preprocessing removes noise from data, handles missing values, and deals with outliers or numbers beyond the typical range of values. The data are subjected to these operations to acquire accurate and superior outcomes from future processing. As a result, cleaning the dataset is necessary before moving on to further activities. This study did data formatting, outlier removal, attribute selection, and handling of missing values to prepare our data for future processing. Data preparation is discussed in the following parts for Chicago, New York, and Los Angeles.

1) Temporal Dataset Creation: This study processes the crime datasets to predict crime daily and hourly. The aim is to develop a time series for recording the temporal components of the data and then use them to train the model, which can predict the number of crimes. Consequently, crime cases are clustered by the hour and day of the month. These numbers are grouped by the hour, each type of crime is analyzed, and the number of crimes of a particular kind that occurred in a specific hour is determined. Another date feature is the temporary rise of five types of criminal behavior. A Python program calculates the number of crime records within a series of date limits to get each record's number of crimes per hour. Moreover, the data

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would be screened based on the type of crime when there is no match between the given crime data. At the end of the time, the dataset has five columns, each containing crime numbers by type. Therefore, a copy of the crime is produced.

2) Normalization of the Dataset: The normalization technique is commonly exploited in data preparation for ML. This study utilized Min–Max normalization to enhance the efficiency of the crime prediction approaches. Normalization is used to alleviate varies by mapping and to set the same size with no information loss. Similarly, normalization is required for most algorithms to run accurate data modeling. This preprocessing step implies a dataset normalization and changes in numerical data: changing 0 to 1 as in (20). Normalization is one of the tools that can be applied to a single column or multiple columns within the same dataset

$$z = \frac{x_i - \min(x)}{\max(x) - \min(x)}. \tag{20}$$

C. State-of-the-Art Prediction Evaluation Measures

Performance metrics comprising mean absolute percentage error (MAPE), mean absolute error (MAE), and root mean square error (RMSE) have been used to assess the effectiveness of prediction algorithms on the dataset. They are also frequently employed in the literature to measure the effectiveness of the prediction model for time series analysis [61], [64], [65].

1) MAPE: MAPE is used in statistics and machine learning algorithms to measure prediction accuracy, trend analysis, and the loss function of prediction models. It is similar to MAE, except observations are normalized before error computation. However, it has a major drawback when values in time series analysis are zero, and division by zero will be undefined. The MAPE error measure is calculated as shown in

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{A_\text{i} - P_\text{i}}{A_\text{i}}\right]^2. \tag{21}$$

2) Mean Absolute Error (MAE): MAE is a risk measure function used in statistics and machine learning to measure the average squared error between the predicted and actual values. MAE calculates the error between paired observations in time series analysis, as shown in (22). The MAE error is not sensitive to outliers by giving less weight to deviating values

$$\text{MAE} = \frac{\sum_{i=1}^{n}|y_\text{i} - x_\text{i}|}{n}. \tag{22}$$

3) RMSE: RMSE is similar to MSE except that the square root of the MSE value is taken to normalize it so that the accuracy of the interpretation of the proposed model can be made easily. Moreover, the RMSE value closer to zero or exactly zero is considered a perfect model. The estimation formula for RMSE is shown in

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(X_\text{i} - Y_\text{i}\right)^2}. \tag{23}$$

TABLE II
HYPERPARAMETER CONFIGURATION AND TUNING STRATEGIES FOR START MODEL

Parameter	Values
Spatiotemporal encoder layers	3
Spatiotemporal decoder stacks	3
Dimensionality of the architecture	512
Hidden layers size	2048
Number of VARA heads	8
Size of the input Kernel	5
K	0, 1, 2, 3,4
Lookback window size	48, 96, 192, 336, 720
Adaptable learning rate	2e-3, 2e-5, 3e-2, 4e-1, 5e-1

D. Results and Discussion

This section details the experimental analysis and evaluation process to assess the efficacy of the proposed approach in predicting crime occurrences according to crime types. This study focuses on the five most critical types of crimes: petit larceny, harassment, burglary, criminal mischief, and grand larceny, from all three datasets.

1) Hyperparameters for Experimental Evaluation: This study adopted a consistent configuration across the proposed START approach consisting of encoder–decoder level and stacks, feed-forward layers, attention mechanism, and the same set of kernel size, as shown in Table II. Hyperparameter tuning focused on three fundamental parameters: k number of frequencies, input window size, and an adaptable learning rate. The grid search strategy was applied on a range of predefined parameters, as outlined in Table II. Particularly, the input window search range was adjusted to the output horizon sizes of the respective datasets to ensure the inclusion of significant historical context. Finally, the parameter combination with significant performance on the validation set using MAE, MAPE, and RMSE scores across three runs was chosen for experimental analysis.

2) Optimization of Proposed Method: This study employed the Adam optimizer due to its ability to capture sparse gradients to optimize the performance of the proposed method [66]. Adam is primarily useful for its adaptive capabilities to capture intrinsic properties. Moreover, the decay rates ($\beta 1 = 0.95$ and $\beta 1 = 0.85$) are selected based on a critical evaluation to capture gradients for ensuring the proposed approach's efficiency and stability. Furthermore, the epsilon parameter is initialized with a suitable value ($\epsilon = 1e - 08$) to avoid division by zero error for enhancing algorithm execution.

The training phase is performed with a batch size of 64 to ensure a balance between computational efficiency and performance. This study employed a learning rate optimization process with linear warmup using the first three epochs to select the optimal rate starting from zero gradually. This process enables the approach to stabilize in the early phase, followed by cosine Annealing, which decreases the learning rate over time utilizing 20 epochs. this helps the model to converge effectively by ensuring that learning will not diminish at zero.

Similarly, the parameters that provide smoothing and damping higher learning rates are designated higher than the base

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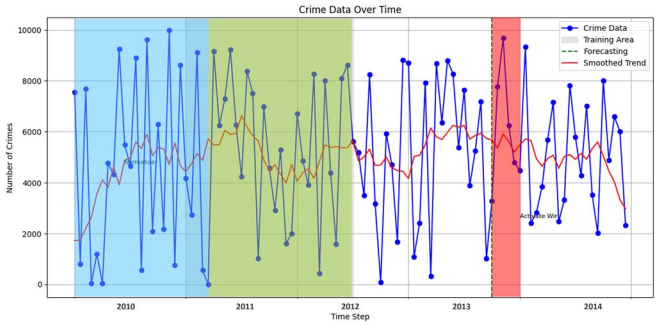

Fig. 4. Training, validation, and testing for time-series forecasting using sliding window.

TABLE III
ABLATION STUDY ON STL DECOMPOSITION

Method	Measure	Chicago	New York	Los Angeles
START W/ STL	MAPE MAE	**0.37** **5.90**	**0.44** **7.50**	**0.40** **6.40**
START W/O STL	MAPE MAE	0.68 6.40	0.90 8.60	0.81 7.32
START W/ VARA	MAPE MAE	0.39 6.10	0.46 7.80	0.42 6.65
START W/O VARA	MAPE MAE	0.76 6.88	0.99 9.05	0.90 7.85

Note: The bold entries show the low error rate in three cities when we use STL decomposition which is core component of proposed model. Ablation studies low score show a significant improvement.

rate to capture intrinsic properties in the data. The proposed approach is implemented on an Intel (R) E5-8770 V3 @ 2.20GHz CPU and an NVIDIA TITAN GPU, providing the necessary computational resources for efficient processing and model training.

3) Training and Testing Dataset: Crime data from Chicago, New York, and Los Angeles cities from 2010 to 2017 are utilized in this study to train and evaluate the model. To train the model, the first four years of data (2010–2013) are used as a training set, and one year of data for testing (80% training and 20% testing). The first four years of data will be used to forecast the next year's crime and so on, as shown in Fig. 4. The sliding window method is used to train the model in different experimental setups. Fig. 4 shows the sliding window strategy. The blue color shows the input window, and the green color shows the output. In addition, the red bar shows the validation set used in the training process. At each step, crime data are normalized and smooth to enhance the capabilities of the proposed method. The following sections discuss the windowing procedure by utilizing STL decomposition at each input.

a) Ablation Study on STL Decomposition: This study investigates the impact of STL decomposition on prediction accuracy. The decomposition allowed the model to isolate nonstationary temporal patterns and focus on residual variations during training. Experimental evaluation in Table III indicates that incorporating STL decomposition improved overall predictive performance across datasets, reducing MAPE, MAE, and RMSE by 10%–15% compared to models without decomposition.

An ablation study was conducted to quantify the effects, comparing the model's performance with and without STL decomposition as shown in Table III. The results demonstrate that STL decomposition significantly enhances predictive accuracy, particularly in datasets with strong seasonal and nonlinear trends. We also compare the case when MH-VARA is replaced with a plain multihead attention and confirm that our trend attention formulation is better.

4) Windowing for Hourly and Daily Crime Predictions Using STL: This study utilized STL decomposition with a sliding window approach for hourly and daily crime prediction across multiple crime types. Crime data from New York, Chicago, and Los Angeles of the top 5 crimes are utilized as input by applying one hot encoding to transform categorical crime type labels into numerical vectors. STL decomposition extracts Seasonal, trend, and remainder components from each crime series. First, an initial analysis is performed to select the appropriate window size (k) for daily and hourly crime predictions considering seasonality. The window size should not exceed the dominant seasonal cycle to avoid incorporating irrelevant information. Therefore, different window sizes are evaluated within the seasonal cycle using MAPE, MAE, and RMSE with execution time.

Therefore, for hourly prediction (t), a window is created utilizing the deseasonalized and detrend crime data according to crime type for the previous k hour (t:k+1 to t), as shown in Fig. 4. This window size is used as input to the proposed model (blue color) to forecast the crime occurrences for the next hour [t+1 (green color)] according to crime type. This process continues by sliding the window to the next hour for the prediction process. Similarly, for daily crime prediction (d), a window is created leveraging deseasonalized and detrend crime data for each crime type for the previous k days (d:k+1 to d). This window size is given as input to the proposed model to forecast the crime occurrences for the next day according to crime type. The process is repeated by sliding the window d+1 for the next day.

5) Experiment of Moving Windows Hourly: This study establishes a range of window sizes (K) for hourly and daily crime predictions. For hourly predictions, k6, k12, k18, and k24 window sizes ensure that the window sizes do not exceed the hourly seasonal cycle. These values show the number of input parameters and are represented by the parameter k. For example, k6 refers to six input time series that are used as input to the proposed model to forecast the next crime (i.e., the seventh crime), while k12 depicts 12 input values of the time series as input values to the input to forecast the next crime.

Table IV shows the experimental results with each window size. It is evident from the results that the increase in window size adds more error and takes more time. Since the sample data span four years, from 2010 to 2013, the projection for the following year, 2014, and the input window for the next three years were chosen accordingly. Window size 6 achieves optimal values as compared to others, as shown in Table IV.

6) Experiment of Moving Window Daily: This section discusses the experimental evaluation of the proposed model with a window size of 24 to 168 for daily predictions for a week using STL decomposition. Daily window sizes aim to capture recurring patterns within the crime data. Window sizes k24,

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TABLE IV
EXPERIMENTAL RESULTS OF MOVING WINDOW HOURLY

Dataset	Measure	k6	k12	k18	k24
Chicago	MAPE	**0.35**	0.41	0.49	0.83
	MAE	**5.83**	6.15	6.86	7.36
	RMSE	**9.15**	10.10	11.53	13.17
	Time (s)	**16**	21	35	48
New York	MAPE	**0.41**	0.48	0.53	0.95
	MAE	**7.13**	7.95	8.56	10.01
	RMSE	**11.47**	12.10	13.21	14.12
	Time (s)	**18**	25	37	56
Los Angeles	MAPE	**0.37**	0.40	0.48	0.85
	MAE	**6.13**	7.05	7.56	8.16
	RMSE	**9.87**	10.69	12.03	14.21
	Time (s)	**17**	23	36	49

Note: Low error rate and time shows optimal results with k6 window size for Hourly predictions.

TABLE V
EXPERIMENTAL RESULTS OF MOVING WINDOW DAILY

Dataset	Measure	k24	k72	k120	k168
Chicago	MAPE	**0.95**	1.41	2.71	3.16
	MAE	**8.76**	10.32	11.04	12.31
	RMSE	**13.21**	14.49	15.98	17.07
	Time (s)	**50**	61	74	89
New York	MAPE	**1.15**	1.98	3.23	4.19
	MAE	**9.13**	10.11	12.96	14.09
	RMSE	**15.43**	16.89	18.32	20.45
	Time (s)	**55**	64	76	92
Los Angeles	MAPE	**0.97**	1.47	2.88	3.55
	MAE	**8.43**	10.07	12.87	13.96
	RMSE	**14.25**	15.13	16.83	18.37
	Time (s)	**52**	60	73	90

Note: Low error rate and time shows optimal results with k6 window size for daily predictions.

k72, k120, and k168 are utilized to choose the optimal window size by ensuring it does not exceed the dominant daily seasonal cycle.

Table V shows k24 window superiority that outperforms the other windows in all evaluation metrics (MAPE, MAE, and RMSE) in all three state-of-the-art datasets. Therefore, the inputs of the window size of k24 are chosen for further analysis.

7) Experiment of Hourly Crime Forecasting: This section presents the experimental analysis of hourly crime forecasting using five types of crimes. Crime data are segmented into hourly windows, and each window contains crime types in each hourly observation. Spatiotemporal features are extracted, and the encoder captures the spatial and temporal relationships that are present in crime occurrences. The spatiotemporal decoder reconstructs the predicted crime occurrence for the next hour by utilizing insights from level and stack. The proposed method (START) is compared with the baseline method TCP [67] and DeepCrime [68] using two alternative approaches. The first approach uses statistics on the total number of crimes without considering the total number of different categories of crimes. The second approach combines the two models with the number of crime types to forecast the overall number of crimes, as shown in Table VI.

It is evident from Table VI that the proposed START method achieves significantly optimal scores compared to baseline methods in two experimental setups. Primarily, START, including crime type, achieves low MAPE, RMSE, and MAE values.

TABLE VI
COMPARATIVE ANALYSIS OF PROPOSED METHOD WITH BASELINE METHODS OF PREDICTING NUMBER OF CRIMES HOURLY WITH AND WITHOUT INCLUDING CRIME

Year	Model	Type	MAPE	MAE	RMSE	R^2
2014	TCP	no	1.25	20.36	25.81	0.43
	DeepCrime		0.89	19.68	22.14	0.55
	START		0.60	18.41	18.43	0.69
	TCP	yes	0.92	19.87	21.37	0.51
	DeepCrime		0.78	17.84	19.17	0.70
	START		**0.43**	**11.21**	**12.91**	**0.83**
2015	TCP	no	1.43	27.64	35.49	0.34
	DeepCrime		0.91	20.59	23.19	0.41
	START		0.63	19.52	29.98	0.62
	TCP	yes	0.88	23.56	32.37	0.44
	DeepCrime		0.71	17.84	19.23	0.65
	START		**0.47**	**11.16**	**13.95**	**0.80**
2016	TCP	no	1.10	26.65	27.23	0.44
	DeepCrime		0.96	24.28	25.51	0.51
	START		0.72	21.91	23.46	0.65
	TCP	yes	0.91	22.53	24.32	0.59
	DeepCrime		0.81	20.84	22.15	0.69
	START		**0.54**	**16.61**	**18.43**	**0.75**
2017	TCP	no	1.95	31.49	35.73	0.25
	DeepCrime		0.89	29.58	32.16	0.36
	START		0.79	27.06	30.14	0.44
	TCP	yes	0.88	25.15	29.19	0.46
	DeepCrime		0.80	24.10	25.84	0.45
	START		**0.67**	**18.36**	**21.09**	**0.70**

Note: Low error rate shows the significance of START model compare to state-of-the-art methods.

In addition, the R^2 value has also been improved in the proposed method compared to others under the same experimental setup. DeepCrime performs better compared to TCP in two experimental setups. Therefore, for crime prediction according to crime types, DeepCrime is compared with the START for hourly crime predictions, utilizing Chicago, New York, and Los Angeles datasets. START achieved low error scores in all measures compared to DeepCrime, as shown in Table VII.

8) Seasonality Hourly: This section compares the hourly seasonality captured by START with and without the number of crime categories. Seasonality varies for each year by combining the data points for each hour of the year, as shown in Fig. 5. Actual and predicted data for each hour of the year are combined and compared to determine the model performance in capturing seasonality in two experimental setups. It is evident from Fig. 5 that the proposed method performs poorly in capturing seasonality hourly without considering crime types.

Therefore, the second experiment is performed by considering crime type and dividing data accordingly to achieve significant results, as shown in Fig. 6. It is evident from the figure that including crime categories enhances the prediction measures to capture seasonality over the four years, with 2014 and 2017 showing better results. In the next part of the evaluation, daily data will be used instead of hourly data to enhance the findings further and confirm that the proposed technique is the best.

This section presents the experimental analysis of hourly crime forecasting using five types of crimes. Crime data are segmented into hourly windows, and each window contains crime types in each hourly observation. Spatiotemporal features are extracted, and the encoder captures the spatial and temporal relationships that are present in crime occurrences.

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TABLE VII
COMPARISON OF HOURLY CRIME PREDICTIONS ACCORDING TO FIVE TYPES OF CRIME FOR CHICAGO, NEW YORK, AND LOS ANGELES USING MAPE, MAE, AND RMSE METRICS WITH THE PROPOSED APPROACH AND STATE-OF-THE-ART METHODS

City	Crime Type	START			DeepCrime		
		MAPE	**MAE**	**RMSE**	MAPE	MAE	RMSE
Chicago	Petit Larceny	0.48	11.25	13.01	0.81	16.24	18.60
	Harassment	0.49	11.35	14.71	0.98	18.81	19.20
	Burglary	0.61	13.70	17.30	1.10	22.50	26.61
	Criminal Mischief	0.54	12.80	15.50	0.97	19.90	28.30
	Grand Larceny	0.58	14.10	17.20	1.18	19.80	26.90
New York	Petit Larceny	0.48	10.90	14.50	0.92	18.20	26.70
	Harassment	0.45	10.70	14.01	0.89	17.80	21.20
	Burglary	0.57	13.40	16.90	0.94	20.50	27.10
	Criminal Mischief	0.50	11.80	15.30	1.60	20.10	27.50
	Grand Larceny	0.53	13.05	16.80	1.70	24.80	30.50
Los Angeles	Petit Larceny	0.46	10.50	14.10	0.86	16.80	21.30
	Harassment	0.43	10.20	13.60	0.85	15.50	23.10
	Burglary	0.55	13.06	16.60	1.05	21.20	28.60
	Criminal Mischief	0.48	11.40	14.90	0.97	19.60	27.10
	Grand Larceny	0.51	12.70	16.40	1.11	19.50	29.20

Note: The bold entries show the low error values when comparing START method in comparing with stat-of-the-art by predicting crime according to crime type.

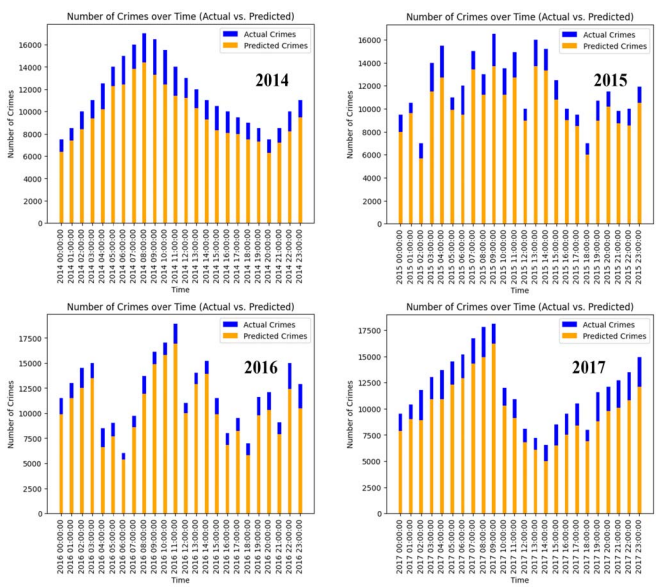

Fig. 5. Without including crime types, the seasonal hourly histogram. The color blue represents actual data, whereas the color orange represents forecasted data.

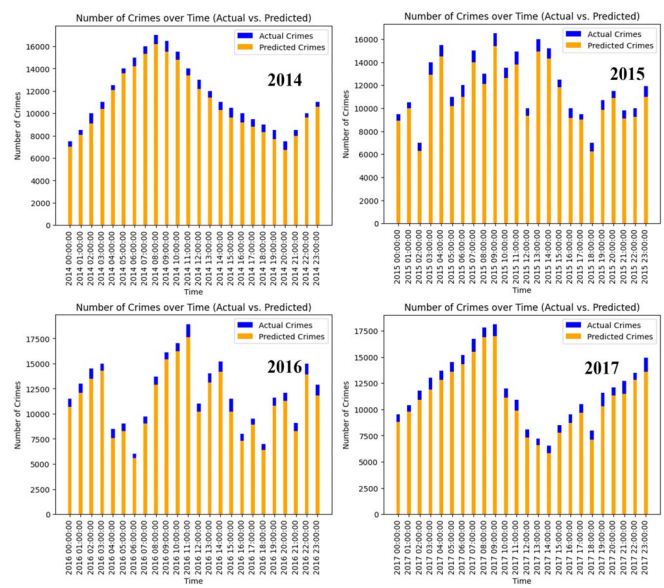

Fig. 6. Histogram seasonal hourly, with crime categories. The orange and blue color lines represent the predicted and actual data, respectively.

The spatiotemporal decoder reconstructs the predicted crime occurrence for the next hour by utilizing insights from level and stack. The proposed method (START) is compared with the baseline method TCP [67] and DeepCrime [68] using two alternative approaches. The first approach uses statistics on the total number of crimes without considering the total number of different categories of crimes. The second approach combines the two models with the number of crime types to forecast the overall number of crimes, as shown in Table VI.

9) Experiment of Daily Crime Forecasting: In this section, daily crime data are used to evaluate the performance of the proposed technique. Crime data are segmented into daily window sizes for a week, and each window contains crime types for daily observation. Spatiotemporal features are extracted, and the encoder captures the spatial and temporal relationships that are present in crime occurrences. The spatiotemporal decoder reconstructs the predicted crime occurrence for the next day by utilizing insights from level and stack. The proposed method (START) is compared with the baseline method TCP [67] and DeepCrime [68] using two alternative approaches. The first approach uses statistics on the total number of crimes without considering the total number of different categories of crimes. The second approach combines the two models with the number of crime types to forecast the overall number of crimes, as shown in Table VIII.

Table VIII demonstrates that incorporating the number of crime types improves the performance of the proposed model compared to the other method. However, Table VI indicates that

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TABLE VIII
COMPARATIVE ANALYSIS OF PROPOSED METHOD WITH BASELINE METHODS OF PREDICTING NUMBER OF CRIMES DAILY WITH AND WITHOUT INCLUDING CRIME

Year	Model	Type	MAPE	MAE	RMSE	R^2
2014	TCP	no	20.71	57.18	86.73	0.23
	DeepCrime		16.20	45.14	63.12	0.30
	START		12.10	32.30	43.50	0.40
	TCP	yes	17.40	49.32	66.84	0.32
	DeepCrime		12.50	35.28	53.80	0.46
	START		**9.12**	**22.20**	**29.10**	**0.55**
2015	TCP	no	19.56	56.90	84.25	0.21
	DeepCrime		17.74	44.60	76.95	0.34
	START		13.10	35.25	54.48	0.45
	TCP	yes	16.18	48.17	74.60	0.39
	DeepCrime		13.60	32.20	53.45	0.55
	START		**10.31**	**25.20**	**33.10**	**0.62**
2016	TCP	no	25.40	62.47	89.05	0.11
	DeepCrime		17.60	51.20	76.20	0.35
	START		14.20	39.40	54.30	0.52
	TCP	yes	21.20	47.30	73.10	0.29
	DeepCrime		14.30	43.40	64.20	0.42
	START		**11.24**	**30.10**	**37.40**	**0.65**
2017	TCP	no	22.50	57.03	76.40	0.19
	DeepCrime		18.90	50.06	65.32	0.35
	START		14.70	42.20	57.06	0.43
	TCP	yes	19.70	49.50	62.39	0.40
	DeepCrime		14.12	38.10	47.30	0.52
	START		**10.30**	**29.09**	**34.20**	**0.67**

Note: The bold entries show the low error rate of START when comparing to state-of-the-art methods with and without including crime type.

the proposed model significantly differs in MAPE, RMSE, and MAE when forecasting hourly data. This is because the first stage of the model, which forecasts the number of crime types in the daily forecast, does not perform well due to the smaller number of samples available for daily data compared to hourly data (35 035 and 1460 for hourly and daily samples from 2010 to 2013, respectively), resulting in fewer crime categories being included in the daily sampling. Moreover, when R-squared was evaluated, the suggested model with crime categories, START, outperformed the other models with values greater than 40%.

In addition, a comparative analysis is performed using the DeepCrime method with START for daily crime prediction according to five types of crimes, utilizing Chicago, New York, and Los Angeles datasets. START achieved low error scores in all measures compared to DeepCrime, as shown in Table IX. However, error values are much higher in daily forecasting compared to hourly. One possible reason would be that hourly predictions can capture more fine-grained details about crime occurrences. For hourly predictions, the model has richer information on how crime unfolds over time compared to daily predictions, where it struggles to capture daily nuances and patterns.

10) Seasonality Daily: This section compares the daily seasonality captured by START with and without the number of crime categories. Seasonality varies by combining the data points for each day of the year, as shown in Fig. 7. Actual and predicted data for each day of the year are combined and compared to determine the model performance in capturing seasonality in two experimental setups. It is evident from Fig. 7 that the proposed method performs poorly in capturing seasonality hourly without considering crime types. The error values in daily predictions are much higher than hourly predictions with and without crime type.

In addition, daily seasonality is calculated by considering crime types, which enhances performance and lowers error values. So, it is evident from all these hourly or daily experiments that predicting crime according to crime type is more beneficial, as shown in Fig. 8.

E. Discussion

The main advantage of the START model is the ability to deal with nonstationary crime data by incorporating STL and VAR attention, which are appropriate for time-varying trends and seasonality. Furthermore, the transformer architecture provides remarkable performance in learning long-range dependencies, allowing the model to generate accurate crime prediction for extended periods. This model offers crime type specific predictions which provide more focused inferences by categorizing differing crime patterns. The model has shown promising scores concerning larger datasets from the major U.S. cities comparing other baseline models. However, START comes with certain limitations. The high data demand may compromise its performance in data-sparse crime regions and computational heaviness resulting from a hybrid approach combining STL, multihead attention, and VAR analysis. An additional limitation to mention is that the model may not generalize well for types of crime without evident seasonality.

Despite this, the practical rewards of what START can do in the real world are immense. The performance improvement of MAPE directly depicts practical advantages for law enforcement. With higher prediction accuracy compared to existing systems, agencies can better identify criminogenic places, which enables the agencies to deploy patrol resources on high-risk areas at the right time. On balance, this reduction in MAPE empowers law enforcement agencies to better plan their resources, hence reducing cost and the overall strategy of preventing crime. It also lays the groundwork for predictive analytics to be used in smart city infrastructures for real-time public safety. The model has the potential to reduce crime rates, restore public trust, and achieve cost efficiencies in preventing-crime efforts by switching law enforcement strategies from reactive to proactive.

Moreover, adapting the proposed model to the diverse urban environment is crucial as variability exists in crime trends, patterns, and data availability across different cities. The proposed approach can be adapted by fine-tuning the hyperparameters to understand the significant crime dynamics across cities. In cities with limited data, transfer learning can be adopted by leveraging patterns learned from large urban environments of similar dynamics to enhance prediction accuracy [69]. In addition, demographic and socioeconomic data incorporation can help the model to capture region-specific patterns efficiently. These adaptation strategies also guarantee consistent reliability and generalizability of the model, irrespective of diverse crime-defined city characteristics and disparities in the availability of data, thereby making them suitable for wider adoption and deployment in different cities.

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TABLE IX
COMPARISON OF DAILY CRIME PREDICTIONS ACCORDING TO FIVE TYPES OF CRIME FOR CHICAGO, NEW YORK, AND LOS ANGELES USING MAPE, MAE, AND RMSE METRICS WITH THE PROPOSED APPROACH AND STATE-OF-THE-ART METHODS

City	Crime Type	START			DeepCrime		
		MAPE	**MAE**	**RMSE**	MAPE	MAE	RMSE
Chicago	Petit Larceny	10.28	25.10	18.20	12.81	49.24	29.59
	Harassment	11.40	27.19	20.60	15.98	58.97	37.59
	Burglary	9.35	19.54	13.30	14.30	29.32	23.60
	Criminal Mischief	12.78	32.56	27.10	18.07	49.76	38.99
	Grand Larceny	11.50	28.75	22.34	17.45	39.26	33.69
New York	Petit Larceny	11.42	27.78	23.39	17.87	38.99	32.76
	Harassment	13.40	37.10	31.21	18.89	47.91	39.19
	Burglary	9.30	24.35	17.95	14.94	40.96	36.20
	Criminal Mischief	12.10	31.65	25.12	17.60	47.63	39.96
	Grand Larceny	12.53	32.56	25.26	21.33	54.10	46.28
Los Angeles	Petit Larceny	10.05	22.67	18.90	16.67	39.98	32.68
	Harassment	12.45	32.54	27.65	18.14	45.22	34.90
	Burglary	8.98	25.87	19.35	14.15	40.12	31.79
	Criminal Mischief	11.04	27.60	19.95	17.67	39.79	32.43
	Grand Larceny	10.46	25.65	18.38	19.24	38.19	30.90

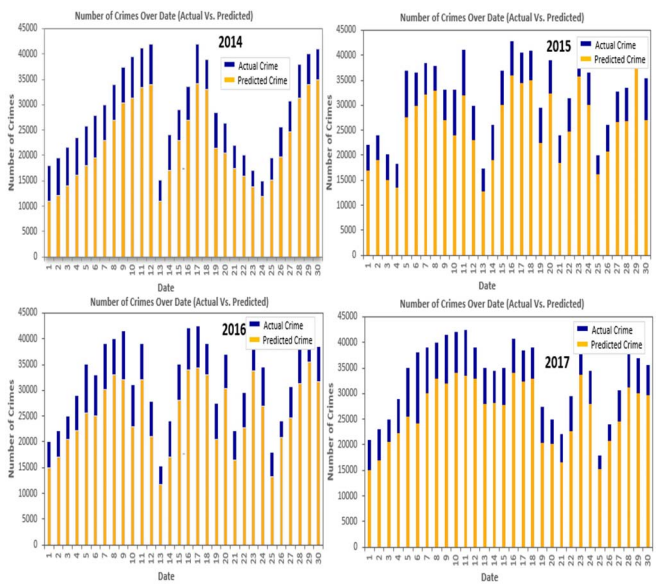

Fig. 7. Seasonal daily histogram excluding crime types actual data are shown by the blue line, while the orange line represents predicted data.

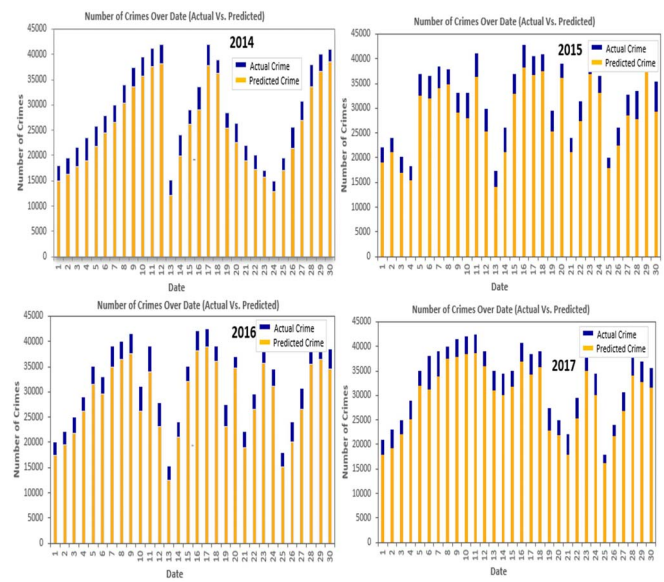

Fig. 8. Histogram Daily, encompassing several sorts of crime. The blue line shows actual data, while the orange line represents predicted data.

F. Comparative Analysis With State-of-the-Art

This section presents a comparative analysis of the proposed START method, a state-of-the-art approach for crime prediction. The performance was evaluated using Chicago, New York, and Los Angeles crime datasets from 2014 to 2017, as shown in Fig. 9. The proposed method was compared with various spatiotemporal techniques, including spatiotemporal sequential hypergraph network (STSHN) [70], spatiotemporal transformer with relative embeddings (STTREs) [37], structured transformer networks (STtrans) [71], and spatiotemporal convolutional transformer network (STCTN) [72], all tested under the same conditions.

For experimental evaluation, hourly crime data are used, as daily fluctuations may be due to random events. Hourly data capture these random seasonal fluctuations, enabling more accurate predictions for the next hour. The quality, computational

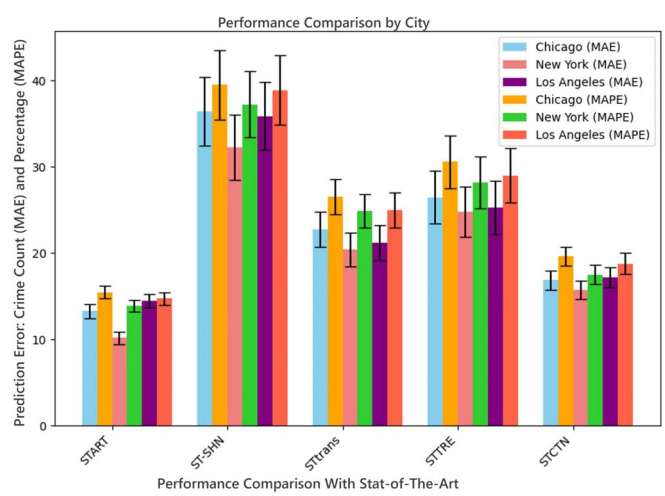

Fig. 9. Comparison of the proposed method with state-of-the-art methods.

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demand, and completeness of hourly data are critical factors compared to daily data. Identical areas across all cities were used to compare with state-of-the-art methods, and hyperparameters were carefully tuned for optimal results. Performance was compared over three years using MAE and MAPE metrics. Black lines on the bars represent the standard deviation of error values for each method and dataset. A consistent assessment window was maintained for each strategy. Experimental results demonstrate the proposed method's superiority, with low MAE (13.30, 10.20, 14.50), MAPE (15.50, 13.90, 14.75), and standard deviation values across three regions, as shown in Fig. 9.

V. Conclusion

This study presents START by leveraging the strengths of spatiotemporal transformer and VAR for tackling time series forecasting. The strength of the proposed model lies in their complementary abilities to capture intricate spatial relationships in the data on a spatial grid and understand temporal dynamics within each location. The proposed model can learn long-range dependencies by identifying potentially significant patterns overlooked by simpler models by modeling the joint dynamics of multiple time series. Primarily, START can capture the nonstationary behavior of data by adapting changes in spatial relationships and evolving temporal dependencies over time.

Moreover, the proposed architecture is based on three fundamental principles. First, it leverages multilevel stacking to extract the latent residual's seasonality, trend, and level representations. Primarily, One hot coding is applied to input data followed by STL decomposition to extract seasonal, trend, and remainder components to help the proposed model capture nonstationary trends and seasonality. Second, the architecture utilizes STL with VAR to extract salient seasonal patterns and trend components by assigning higher weight to more recent components. In addition, VAR is utilized for multivariate analysis and as an attention mechanism by predicting crime according to crime types. Finally, the prediction is a composition of seasonal, levels, and trend components. The proposed model outperforms state-of-the-art algorithms with low MAE and MAPE scores.

However, START needs significant training data, and the hybridization requires careful parameter integration. Future research can integrate social and demographic data into the model to capture more holistic information on crime patterns, improving crime prediction accuracy.

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